

GTech Academy Networking Curriculum(starter-pack)

The Story of Networking: From Simple Connections to Global Networks

Learning Philosophy — The GTech Method

Every networking concept will be introduced through a real engineering problem.

Students will always answer:

1. **What problem existed?**
2. **Why were previous methods not enough?**
3. **What solution did engineers create?**
4. **How does the technology work?**
5. **How do we implement it practically?**
6. **What new problems does this solution create?**

Each stage includes:

- Historical Context
 - Engineering Problem
 - Previous Limitations
 - New Technology Solution
 - Technical Concepts
 - Packet Tracer Laboratory
 - Future Challenges
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Stage 1 — The Beginning of Computer Communication

The World at the Time

Early computers operated independently.

Organizations had multiple computers, but each machine worked as an isolated system.

Sharing information required physical methods such as:

- USB drives
- CDs
- Manual file transfer

Computers needed a way to communicate directly.

The Problem

Computers could process information, but they could not exchange data efficiently.

Organizations needed:

- Faster communication
 - Resource sharing
 - Centralized access to information
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Why Existing Methods Failed

Manual data transfer was:

- Slow
- Inefficient
- Difficult to manage
- Not suitable for growing organizations

Engineers needed a way for computers to communicate automatically.

The New Solution

Computer Networks.

Devices were connected using:

- Network Interface Cards (NICs)
 - Cables
 - Network devices
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Students Learn

- What a computer network is
 - Types of networks
 - LAN concepts
 - Network devices
 - NICs
 - MAC addresses
 - Basic communication between devices
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Hands-on Lab

Packet Tracer:

- Connect multiple computers
 - Configure a basic LAN
 - Test connectivity using ping
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New Limitation

"Computers can now communicate, but how does each device know where to send information?"

This creates the need for addressing.

Stage 2 — Giving Computers Identity: Static IP Addressing

The World at the Time

Networks were growing.

More computers were being connected together.

Each device needed a unique identity.

The Problem

Computers connected to the network needed addresses.

Without addresses:

- Devices could not locate each other
 - Data could not reach the correct destination
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Why Existing Methods Failed

There was no automatic way to identify devices.

Network administrators had to manually configure:

- IP address
 - Subnet mask
 - Default gateway
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The New Solution

Static IP Addressing.

Administrators manually assigned network addresses to every device.

Students Learn

- IPv4 addressing
 - Subnet masks
 - Default gateways
 - Private IP addresses
 - Network and host portions
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Hands-on Lab

Packet Tracer:

- Configure static IP addresses
 - Create a small LAN
 - Test communication
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New Limitation

"Static addressing works for a small network, but what happens when an organization has hundreds or thousands of devices?"

Stage 3 — Automatic Addressing: DHCP

The World at the Time

Organizations started expanding.

Schools, companies, and institutions had hundreds of computers.

The Problem

Manually configuring every device became impossible.

Imagine configuring:

- 500 computers

- 5,000 computers
 - Multiple branches
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Why Existing Methods Failed

Static IP addressing created:

- Human errors
 - IP conflicts
 - Slow deployment
 - High administrative workload
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The New Solution

Dynamic Host Configuration Protocol (DHCP).

DHCP automatically provides:

- IP address
 - Subnet mask
 - Default gateway
 - DNS information
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Students Learn

- DHCP process (DORA)
 - DHCP server
 - DHCP clients
 - Address leasing
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Hands-on Lab

Packet Tracer:

- Configure a DHCP server
 - Connect multiple clients
 - Observe automatic addressing
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New Limitation

"The network can now support many devices, but what happens when one network becomes too large?"

Stage 4 — Separating Networks: The Birth of Routing

The World at the Time

Organizations continued growing.

A single LAN could contain hundreds of devices.

The Problem

Large networks created too much traffic.

Every broadcast reached every device.

Network performance decreased.

Why Existing Methods Failed

One large network caused:

- Broadcast storms
 - Poor performance
 - Difficult management
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The New Solution

Routers.

Routers connect different networks and control traffic flow.

Students Learn

- Why routers exist
 - Network segmentation
 - Broadcast domains
 - Routing tables
 - Default gateway
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Hands-on Lab

Packet Tracer:

- Connect two networks using a router
 - Configure interfaces
 - Test communication
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New Limitation

"Routers can communicate, but how do they know where other networks exist?"

Stage 5 — Manual Route Management: Static Routing

The World at the Time

Organizations now had multiple networks connected through routers.

The Problem

Routers needed information about remote networks.

Why Existing Methods Failed

Routers did not automatically know:

- Where networks were located
- Which path to use

Administrators had to manually add routes.

The New Solution

Static Routing.

Network engineers manually create routing entries.

Students Learn

- Routing tables
 - Static routes
 - Next-hop addresses
 - Path selection
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Hands-on Lab

Packet Tracer:

- Connect three routers
 - Configure static routes
 - Verify routing tables
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New Limitation

"Static routing works for small networks, but would you configure 150 routers manually?"

Stage 6 — Learning Routes Automatically: Dynamic Routing (RIP)

The World at the Time

Large companies began operating many branches.

Networks contained many routers.

The Problem

Manual routing became impossible.

Every change required administrators to update multiple routers.

Why Existing Methods Failed

Static routing could not handle:

- Large networks
- Frequent changes
- Link failures

The New Solution

Dynamic Routing Protocols.

Routers communicate with each other and automatically learn routes.

First solution:

Routing Information Protocol (RIP)

Students Learn

- Dynamic routing concepts
 - Routing protocols
 - RIP operation
 - Hop count metric
 - Routing updates
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Hands-on Lab

Packet Tracer:

- Build a three-router network
 - Configure RIP
 - Observe automatic route learning
 - Remove a route and observe adaptation
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New Limitation

RIP has weaknesses:

- Maximum 15 hops
 - Slow convergence
 - Not suitable for very large networks
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Stage 7 — Controlling Broadcast Traffic: VLANs

The World at the Time

Companies placed hundreds of users on switches.

The Problem

Large LANs created excessive broadcast traffic.

Why Existing Methods Failed

A single broadcast domain was inefficient.

The New Solution

Virtual Local Area Networks (VLANs).

Students Learn

- VLAN concepts
 - Access ports
 - Trunk ports
 - Broadcast domains
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Hands-on Lab

Create:

- Multiple VLANs
 - Department separation
 - VLAN assignments
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New Limitation

"Departments are separated, but how do different VLANs communicate?"

Stage 8 — Connecting VLANs: Inter-VLAN Routing

Problem

VLANs improved security and performance but isolated departments.

Solution

Inter-VLAN Routing.

Students Learn

- Router-on-a-stick
 - Layer 3 switching
 - VLAN gateways
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Lab

Configure communication between VLANs.

Stage 9 — DHCP Across Different Networks

Problem

DHCP broadcasts cannot cross routers.

Clients in different VLANs cannot reach the DHCP server.

Solution

DHCP Relay.

Students Learn

- DHCP relay
- ip helper-address
- Routed DHCP communication

Lab

Configure DHCP relay between VLANs.

Stage 10 — Enterprise Routing: OSPF

The Problem

RIP cannot support modern enterprise networks.

Organizations need:

- Faster convergence
 - Better scalability
 - Intelligent path selection
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The Solution

Open Shortest Path First (OSPF).

Students Learn

- Link-state routing
 - Areas
 - Cost metrics
 - Neighbor relationships
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Lab

Configure multi-router OSPF networks.

Stage 11 — Connecting the World: BGP

The World at the Time

The Internet became a network of thousands of organizations.

The Problem

Different organizations needed to exchange routing information.

A single company's routing protocol could not control the entire Internet.

The Solution

Border Gateway Protocol (BGP).

Students Learn

- Autonomous Systems
 - Internet routing
 - External routing
 - How ISPs communicate
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Final Lab Project

Students design and build:

"A Small Enterprise Network Connected to the Internet"

Including:

- VLANs
 - DHCP
 - Routing
 - OSPF
 - Security concepts
 - Internet connectivity concepts
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Final Outcome

By the end of this journey, students will not only know networking technologies, but understand:

Why networks evolved.